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Lualaba/Kolwezi

Senior Data Engineer – Mining operations

Professional Summary

Results-driven Software and Data Engineer with 15+ years of experience in designing and building software solutions, and enterprise data infrastructure. Expertise in developing scalable data architecture, data pipelines, optimizing databases, and implementing data-driven solutions with hands-on experience in cloud computing, machine learning, and big data processing. Adept at collaborating with stakeholders to architect and deploy data solutions that drive strategic decision-making.

Technical Skills

- **Data Engineering:** ETL/ELT pipelines, data modeling, data warehousing, data lakes
 - **Data orchestration:** Dagster, OpenHIM, Apache airflow
 - **Programming:** Python (FastAPI, Pandas, Numpy, PySpark), SQL, DBT
 - **Databases:** SQL Server, PostgreSQL, MySQL, NoSQL (MongoDB, Elasticsearch)
 - **Big Data & Cloud:** AWS (S3, Lambda, RDS, Redshift, Athena, Glue), BigQuery, Apache Kafka, Apache Spark
 - **DevOps & Automation :** Docker, CI/CD (GitHub Actions)
 - **Project Management :** Agile (Scrum), Jira
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Professional Experience

Senior Data Engineer| Kamoto Copper Company (*Sep 2025 – Present*)

Keywords: Data Engineering, Data warehouse, Data pipelines, Data governance, Data strategy

- Design company data strategy and governance framework
- Design and build modern on-premises company data warehouse (Medallion architecture)
- Design modern ETL/ELT data pipelines that ingest data from multiple sources (Excel, PDF, Database, ERP, SCADA)
- Implement pipelines orchestration and data validation layer
- Work with stakeholders to identify data sources and data need

- Support analytics team with golden data for insights and decision making

Technical Assistant | BlueSquare (Sep 2024 – Present)

Keywords: Data Engineering, Digital Health, Software engineering, Cloud Computing

- Design and build data interoperability layer for the Health Information Exchange
- Design realtime data processing for transfer-in/transfer-out notification
- Design and implement data pipelines to support the integration of health data from multiple sources.
- Design and implement microservices for data processing and integration
- Work with stakeholders to define data architecture for large-scale health projects.
- Support data governance and security compliance.
- Manage different software development projects in the dev team.

Digital Advisor | iplus Solutions (PROSANI USAID) (Dec 2023 – Sep 2024)

Keywords: Data Engineering, Digital Health, Supply Chain Analytics

- Developed scalable ETL pipelines for real-time health supply chain data processing.
- Integrated health data from Medexis with national health systems.
- Designed and deployed cloud-based storage solutions for secure data access.

Data Interoperability & ML Engineer | Pendulum (Remote) (Apr 2022 – Jul 2023)

Keywords: Python, FastAPI, PostgreSQL, AWS, DHIS2, OpenLMIS, mSupply

- Architected an interoperability layer connecting DHIS2, OpenLMIS, and mSupply.
- Build modern ETL/ELT pipelines to extract data from different health information systems
- Built automated data pipelines for ML-based forecasting in health logistics.
- Optimized data storage and retrieval processes for performance and scalability.

Regional Health Software Engineer | IntraHealth (Remote) (Jul 2021 – Apr 2022)

Keywords: FHIR, DHIS2, AWS, PostgreSQL, Keycloak, Elasticsearch

- Build data repository for HIV project accross different projects
- Build data pipeline for integration into the central PEPFAR data repository
- Developed FHIR-based APIs to facilitate interoperability between health information systems.
- Develop Human Resource software accross countries and projects

Health Information Systems Developer | IntraHealth (*Sep 2018 – Jul 2021*)

Keywords: Node.js, Vue.js, PostgreSQL, AWS, ElasticSearch

- Designed and implemented middleware for DHIS2-DATIM/PEPFAR data exchange.
- Developed cloud-based health data analytics dashboards.
- Built and maintained a high-availability data warehouse for HIV program data.

Health Information Systems Developer | IntraHealth (*Nov 2014 – Sep 2018*)

Keywords: PHP, XML, JavaScript, MySQL, AWS, Ubuntu

- Developed an iHRIS-based national health workforce registry.
- Built ETL pipelines to integrate workforce data with national health systems.
- Conducted WISN workload analysis for staffing optimization.

Software Developer | Microcom (*Jan 2011 – Nov 2014*)

Keywords : PHP, Java, MySQL, Apache

- Led the digitization of company processes.
- Develop flagship company information systems
- Design and Developed company databases

Education

- **Computer Information Systems Engineer**, Ecole Supérieure d'Informatique Salama (ESIS), 2010
- Certifications: Data engineering using AWS, SQL Associate Certificate, Associate Data Engineer
- PostGraduate diploma in Data Science: University of Texas, Austin (via greatlearning)

Languages & Interests

- Languages: English, French